

MATH — Unit M1: Functions, Limits & Continuity

Sessions 1-2 (4 hours total) | Track: Mathematical Analysis

Learning Objectives

By the end of this unit, the student should be able to:

- Define a function formally and identify domain/range.
- Compute limits algebraically and understand the ε - δ intuition.
- Determine continuity of a function at a point and over an interval.
- Recognize and evaluate common indeterminate forms.

1. Functions — the building block

A **function** f from set A to set B , written:

$$f : A \rightarrow B$$

assigns to every element $x \in A$ exactly one element $f(x) \in B$.

Domain: the set of all valid inputs, $\text{dom}(f)$.

Range: the set of all outputs actually produced, $\{f(x) : x \in \text{dom}(f)\}$.

Example 1.1 $f(x) = 1/(x - 2)$

Domain: all real numbers except $x = 2$, written $\text{dom}(f) = \mathbb{R} \setminus \{2\}$.

2. The idea of a limit

We write:

$$\lim_{(x \rightarrow a)} f(x) = L$$

to mean: as x gets arbitrarily close to a (but not equal to a), $f(x)$ gets arbitrarily close to L .

Formal (ε - δ) definition

For every $\varepsilon > 0$, there exists a $\delta > 0$ such that:

$$0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$$

You don't need to prove things with ε - δ yet — just understand: *no matter how small a "closeness" ε you demand for the output, we can find a small enough "closeness" δ in the input that guarantees it.*

Left-hand and right-hand limits

$\lim_{(x \rightarrow a^-)} f(x)$ — approaching from below $\lim_{(x \rightarrow a^+)} f(x)$ — approaching from above

A two-sided limit exists **only if** both one-sided limits exist and are equal:

$$\lim_{(x \rightarrow a^-)} f(x) = \lim_{(x \rightarrow a^+)} f(x) = L \Rightarrow \lim_{(x \rightarrow a)} f(x) = L$$

3. Limit laws (the toolkit)

If $\lim_{(x \rightarrow a)} f(x) = L$ and $\lim_{(x \rightarrow a)} g(x) = M$, then:

Rule	Statement
Sum	$\lim [f(x) + g(x)] = L + M$
Product	$\lim [f(x) \cdot g(x)] = L \cdot M$
Quotient	$\lim [f(x)/g(x)] = L/M$, provided $M \neq 0$
Constant multiple	$\lim [c \cdot f(x)] = c \cdot L$
Power	$\lim [f(x)^n] = L^n$

Example 3.1 — Direct substitution $\lim (x \rightarrow 3) (x^2 + 2x - 1) = 3^2 + 2(3) - 1 = 9 + 6 - 1 = 14$

Example 3.2 — Indeterminate form $0/0$, factor and cancel $\lim (x \rightarrow 2) (x^2 - 4)/(x - 2)$

Direct substitution gives $0/0$ (indeterminate — not automatically undefined, needs more work).

Factor: $x^2 - 4 = (x - 2)(x + 2)$

$(x^2 - 4)/(x - 2) = (x - 2)(x + 2)/(x - 2) = x + 2$ for $x \neq 2$

So: $\lim (x \rightarrow 2) (x^2 - 4)/(x - 2) = \lim (x \rightarrow 2) (x + 2) = 4$

Example 3.3 — Rationalizing $\lim (x \rightarrow 0) (\sqrt{x+1} - 1)/x$

Multiply by the conjugate $(\sqrt{x+1} + 1)/(\sqrt{x+1} + 1)$:

$= [(x+1) - 1] / [x(\sqrt{x+1} + 1)] = x / [x(\sqrt{x+1} + 1)] = 1 / (\sqrt{x+1} + 1)$ for $x \neq 0$

$\lim (x \rightarrow 0) 1/(\sqrt{x+1} + 1) = 1/(1 + 1) = 1/2$

4. Continuity

f is **continuous at $x = a$** if all three conditions hold:

$f(a)$ is defined

$\lim (x \rightarrow a) f(x)$ exists

$\lim (x \rightarrow a) f(x) = f(a)$

If f is continuous at every point of an interval, we say f is **continuous on that interval**.

Example 4.1 $g(x) = \{ x + 1, \text{ if } x < 1; 3, \text{ if } x = 1; x^2 + 1, \text{ if } x > 1 \}$

Check continuity at $x = 1$:

$\lim (x \rightarrow 1^-) g(x) = 1 + 1 = 2$

$\lim (x \rightarrow 1^+) g(x) = 1^2 + 1 = 2$

Two-sided limit = 2

But $g(1) = 3$

Since $\lim (x \rightarrow 1) g(x) = 2 \neq g(1) = 3$, **g is NOT continuous at $x = 1$** (a "removable discontinuity").

5. In-Session Guided Practice

Work through these together — tutor should let the student attempt first, then correct.

Find the domain of $f(x) = \sqrt{x - 5}$.

Evaluate $\lim (x \rightarrow 4) (x^2 - 3x + 2)$.

Evaluate $\lim (x \rightarrow 5) (x^2 - 25)/(x - 5)$ by factoring.

Evaluate $\lim (x \rightarrow 0) (x^2 + 3x)/x$.

Is $h(x) = 1/x$ continuous at $x = 0$? Justify using the 3-part definition.

6. Take-Home Questions

Q1. Find the domain and range of $f(x) = 1/(x + 3)$.

Q2. Evaluate: $\lim_{(x \rightarrow 2)} (x^3 - 8)/(x - 2)$ (Hint: $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$)

Q3. Evaluate: $\lim_{(x \rightarrow 9)} (\sqrt{x} - 3)/(x - 9)$ (Hint: rationalize using conjugate $\sqrt{x} + 3$)

Q4. For the piecewise function: $f(x) = \{ 2x + 1, \text{ if } x \leq 2 ; x^2, \text{ if } x > 2 \}$ determine whether f is continuous at $x = 2$.

Q5. Explain in your own words why $\lim_{(x \rightarrow 0)} \sin(x)/x = 1$ cannot be evaluated by direct substitution, and what "indeterminate form" means.

Answer Key (for tutor reference — don't share until reviewed)

$\text{dom}(f) = \mathbb{R} \setminus \{-3\}$, $\text{range} = \mathbb{R} \setminus \{0\}$

Factor: $(x-2)(x^2+2x+4)/(x-2) = x^2+2x+4 \rightarrow \text{at } x=2: 4+4+4 = \mathbf{12}$

Rationalize $\rightarrow 1/(\sqrt{x}+3) \rightarrow \text{at } x=9: 1/6$

$\lim_{(x \rightarrow 2^-)} = 5$, $\lim_{(x \rightarrow 2^+)} = 4 \rightarrow \text{limit does not exist} \rightarrow \mathbf{\text{discontinuous at } x=2}$

Direct substitution gives $0/0$, an indeterminate form — meaning the limit isn't automatically 0, undefined, or any fixed value; it must be resolved via algebra, series expansion, or L'Hôpital's rule (covered in Unit M2).